PREDICTIVE MICRO-CLIMATE SUPPORT FOR OYSTER MUSHROOM CULTIVATION USING A TEMPERATURE-HUMIDITY BASED TRANSPIRATION STRESS INDEX (TSI)

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

I declare that this individual thesis describes my own research and development work for the temperature- and humidity-based Transpiration Stress Index (TSI) component of project R26-ISE-002. Material taken from research papers, official sources, the group research paper, and other published work is acknowledged through citations. I understand that I am responsible for the accuracy of the calculations, results, source code, figures, tables, and conclusions included in this thesis.

I also confirm that any use of generative AI or other automated tools during drafting will be disclosed according to the university and module requirements. Any AI-assisted text must be checked against the original research evidence before final submission.

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ABSTRACT

Oyster mushrooms need suitable temperature and humidity throughout cultivation. In Sri Lankan mushroom rooms, these conditions can change during the day, and a farmer may notice the problem only after the environment has already become stressful. Many monitoring systems show temperature and relative humidity as two separate values or use fixed limits. This individual research develops a simple decision-support component that combines both measurements through Vapour Pressure Deficit (VPD) and a project-defined Transpiration Stress Index (TSI). Temperature and relative humidity are measured using a DHT22 sensor and sent through an ESP32-based IoT node to the backend. The system calculates VPD, converts it into a TSI value, checks the recent TSI trend, and presents the result as a room-wise warning and practical recommendation on the farmer's mobile interface. The group prototype uses TSI greater than 65 as the high-stress safety trigger. The exact boundary between low and moderate stress is kept configurable because it still requires field calibration. Reported group evaluation results for the environmental subsystem show temperature accuracy of approximately ±0.3 °C, relative-humidity accuracy of ±2.1%, and an average actuation response of 4.2 seconds. Temperature showed a strong positive relationship with TSI ($r = 0.92$), while relative humidity showed a strong negative relationship ($r = -0.99$). User testing for automation and stress control achieved a mean score of 4.73/5 and 94.6% satisfaction. The main contribution of this thesis is the conversion of raw environmental readings into an understandable moisture-stress warning and a clear next step for the farmer under Sri Lankan cultivation conditions.

Keywords: oyster mushroom, temperature, relative humidity, Vapour Pressure Deficit, Transpiration Stress Index, IoT, early warning, farmer recommendation

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
ESP32	Espressif ESP32 microcontroller
IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
RH	Relative Humidity
TSI	Transpiration Stress Index (project-defined moisture-stress index)
UAT	User Acceptance Testing
VPD	Vapour Pressure Deficit

1 INTRODUCTION

1.1 Overall project context

Oyster mushroom cultivation is an important small-scale agricultural activity because it can be carried out in a limited space and can produce food within a relatively short period. However, mushroom production is sensitive to the surrounding environment. Temperature, humidity, air exchange, disease, and the correct harvesting time all affect the final quality and yield. In many farms, these conditions are still checked manually. A farmer may enter the growing area, look at the mushrooms, feel the air, check a basic meter, and then decide whether to mist or ventilate. This approach depends heavily on experience and may not identify a problem early enough.

The overall R26-ISE-002 research project, titled “Rack-Level Micro-Climate and Growth Optimizer for Oyster Mushroom Farms”, was designed to reduce this dependence on manual observation. The group system contains four main research components: environmental moisture-stress monitoring using TSI, image-based disease detection, growth-stage and harvest-time estimation, and CO2-based stress analysis. These components share one mobile and backend platform, but each component solves a different research problem.

This thesis does not describe all four components as if they were one individual contribution. It focuses only on the first component: using temperature and relative humidity to understand moisture stress, provide an early warning, and give the farmer a clear recommendation. Disease detection, growth-stage classification and CO2 modelling are mentioned only when their integration with the shared platform needs to be explained.

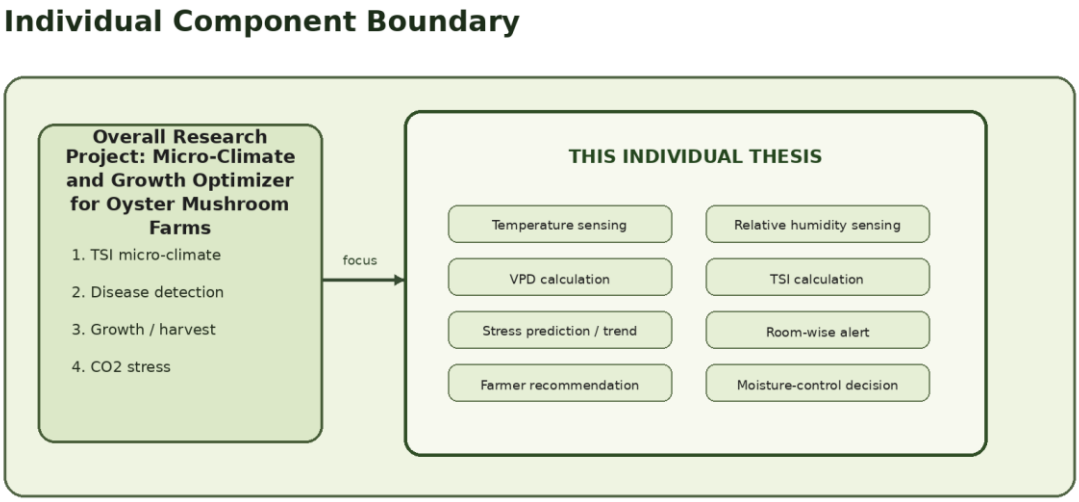


Figure 1.1 Individual component boundary within the overall project.

1.2 Individual component context

The individual component starts with two simple measurements: air temperature and relative humidity. These values are easy to measure, but they are not useful enough when they are shown separately. A temperature value can be acceptable at one humidity level and stressful at another. In the same way, a humidity value cannot be judged properly without considering the temperature at the same time. The aim of this component is therefore to convert the two readings into one environmental stress signal that is easier to understand and act on.

The component uses Vapour Pressure Deficit (VPD) as the physical link between temperature and humidity. VPD expresses how strongly the air can pull moisture from a wet surface. In the context of this project, VPD is then normalized into a Transpiration Stress Index (TSI). The term TSI is retained because it is the name used in the group research. Oyster mushrooms do not transpire through stomata in the same way as plants; therefore, in this thesis TSI should be understood as a project-defined atmospheric moisture-stress index for mushroom cultivation, not as a claim that fungal physiology is identical to plant transpiration.

The useful output is not an LCD value. The farmer receives the result through the mobile system. The component shows the cultivation room, the current temperature, humidity, TSI level, recent trend and a short recommendation. For

example, the farmer may be told to maintain the current condition, inspect the room because stress is rising, increase humidity, or reduce excessive temperature. This makes the component a decision-support tool rather than only a sensor display.

1.3 Problem statement

The main problem is that farmers can receive raw temperature and humidity readings without receiving a clear answer to the practical question: “Is this room becoming stressful for the mushrooms, and what should I do next?” Fixed temperature-only or humidity-only limits can miss the combined effect of both variables. In addition, a reactive system warns only after a limit is crossed, while the farmer needs an earlier indication that conditions are moving in the wrong direction.

This problem is important in Sri Lanka because the recommended environment changes with oyster mushroom variety and growth stage. The Sri Lankan Department of Agriculture lists different temperature and humidity requirements for American oyster, abalone and pink oyster, and the requirements also differ between spawn run, primordia formation and fruiting. This means that one fixed temperature or humidity value is not enough for all cultivation situations.

The individual research problem can therefore be stated as follows: How can temperature and relative humidity data from an oyster mushroom cultivation room be combined into a simple, explainable stress index that can give an early warning and practical farmer recommendation under Sri Lankan cultivation conditions?

1.4 Research gap

Previous IoT-based mushroom systems show that continuous temperature and humidity monitoring is practical. Some systems also switch fans, heaters, pumps or humidifiers when a fixed limit is reached. These solutions reduce manual work, but many of them still treat temperature and humidity as separate control values. They are mainly reactive: a device turns on after the measured value crosses a predefined threshold.

The literature also shows a gap between sensing and farmer decision support. A farmer may see “28 °C” and “82% RH”, but the system may not explain whether

the combined condition is safe, getting worse, or already critical. Several systems also use an LCD located at the device, which is useful for local testing but less useful for a farmer who is away from the cultivation room.

The gap addressed by this individual component is therefore the lack of a simple combined moisture-stress interpretation that connects temperature and humidity to a clear room-wise warning, recent-trend prediction and farmer action. The research does not claim that TSI is a universal biological law for all oyster mushroom varieties. Instead, it investigates TSI as a practical decision-support index that can be calibrated to local cultivation conditions.

1.5 Aim and objectives

The main aim of this research is to design, implement and evaluate a temperature- and humidity-based TSI component that can identify developing moisture stress in oyster mushroom cultivation rooms and provide a clear recommendation to the farmer.

1. To collect continuous temperature and relative humidity readings from the mushroom cultivation environment using an IoT sensor node.
2. To calculate saturated vapour pressure, VPD and a normalized TSI value from the environmental readings.
3. To interpret current TSI and recent TSI change as low, moderate or high stress for farmer communication, while keeping unvalidated boundaries configurable.
4. To provide an early warning when the TSI trend is moving toward the high-stress safety limit.
5. To present room-wise status, warning and recommended next action through the mobile application instead of depending on an LCD.
6. To evaluate sensor accuracy, response time, relationships between temperature/RH and TSI, and farmer acceptance of the stress-control feature.

1.6 Research questions

1. How effectively can temperature and relative humidity be combined through VPD and TSI to describe changes in the cultivation-room moisture environment?
2. What relationship exists between temperature, relative humidity and the calculated TSI in the collected environmental data?
3. Can recent TSI behaviour be used to give a useful early warning before the high-stress safety limit is crossed?
4. Can a room-wise mobile warning and recommendation make environmental information easier for farmers to understand and use?

1.7 Scope, limitations and assumptions

The scope of this thesis includes temperature sensing, relative-humidity sensing, VPD calculation, TSI calculation, short-term trend interpretation, stress-level communication, farmer recommendations, and integration with the shared mobile/backend system. The farmer-facing view is presented by mushroom cultivation room. One room may contain one or more sensor nodes depending on size and layout.

The thesis does not include image-based disease detection, CNN growth-stage classification, harvest-time estimation, or CO2 stress modelling as individual research contributions. Those belong to other members of the overall project. CO2 information may appear in the shared system, but it is not an input to the temperature-humidity TSI calculation described here.

The group paper defines TSI greater than 65 as the safety trigger. It does not provide a validated numeric boundary between low and moderate stress. For this reason, this thesis does not invent one. The low-to-moderate warning boundary is treated as a configurable value that should be calibrated using additional field data. The airflow term in the group TSI formula is also treated as a configured empirical factor unless an airflow sensor is available; temperature and RH remain the live measured inputs of this individual component.

1.8 Individual contribution

The individual contribution is the environmental decision layer that converts raw temperature and relative-humidity telemetry into VPD, TSI, a stress interpretation, an early warning, and a practical farmer recommendation. The contribution also includes the design of room-wise alert messages, the mapping of stress conditions to farmer actions, and the evaluation of the environmental subsystem. The mobile application, shared database and overall architecture are group artefacts; this thesis documents only the parts that directly support or integrate the TSI component.

1.9 Thesis structure

Chapter 2 reviews the environmental needs of oyster mushrooms, VPD, IoT monitoring and related systems. Chapter 3 explains the research method, data collection, TSI calculation, prediction logic and evaluation measures. Chapter 4 presents the product and solution design. Chapter 5 describes implementation, integration and testing. Chapter 6 reports the available results. Chapter 7 discusses what those results mean and compares them with previous work. Chapter 8 considers practical value, scalability and sustainability. Chapter 9 concludes the thesis and identifies future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This review focuses on work that directly supports the individual component. The main topics are oyster mushroom environmental requirements, the combined effect of temperature and humidity, VPD, IoT-based temperature and humidity monitoring, automatic environmental control, and farmer-facing decision support. Papers that focus mainly on disease detection, image classification or robotic harvesting are outside the main review because they do not answer the research question of this thesis.

Sources were selected from the references used in the group research paper, the temperature-and-humidity mushroom paper provided during this research, Sri Lankan publications, and the official Sri Lankan Department of Agriculture mushroom guidance. The review gives more weight to work that explains how environmental values are measured, combined or used for decisions. It also distinguishes between systems that only display sensor readings and systems that turn the readings into an action.

2.2 Oyster mushroom micro-climate and Sri Lankan context

Oyster mushrooms are grown in a moist environment because the fruiting body contains a high proportion of water and can lose surface moisture when the surrounding air becomes too dry. Temperature also affects development rate and the amount of moisture the air can hold. As a result, temperature and humidity need to be considered together during crop management.

Sri Lankan conditions make this especially important because oyster mushroom cultivation is carried out using more than one released variety. The Department of Agriculture provides separate environmental ranges for American oyster, abalone and pink oyster. It also separates the spawn-run stage from primordia and fruiting. Table 2.1 summarizes the official temperature and humidity guidance that is most relevant to this thesis. These values show why a single universal threshold should not be assumed for every room and every stage.

The group research paper used a broad environmental test band of approximately 20–26 °C and 80–90% RH during integrated evaluation. That band is useful for system testing, but it should not replace the stage- and variety-specific values published by the Department of Agriculture. A strong practical version of the TSI component should therefore allow the farmer or administrator to select the variety and growth stage so that the warning logic can use the correct local context.

Variety / stage	Temperature (°C)	Relative humidity (%)
American oyster – spawn run	24–27	90–100
American oyster – primordia	17–19	95
American oyster – fruiting	19–20	85–92
Abalone – spawn run	24–30	90–95
Abalone – primordia	18–24	95–100
Abalone – fruiting	21–27	85–90
Pink oyster – spawn run	24–27	90–100
Pink oyster – primordia	17–19	95
Pink oyster – fruiting	19–20	85–92

Table 2.1 Sri Lankan Department of Agriculture climatic requirements for selected oyster mushroom varieties and stages [11].

2.3 Temperature, relative humidity, VPD and moisture stress

Relative humidity alone can be misleading because the moisture-holding capacity of air changes with temperature. VPD provides a better description of atmospheric drying demand because it uses both temperature and RH. When temperature increases while RH decreases, VPD rises. This means the air has a stronger capacity to remove moisture from a wet surface. The group research uses this physical relationship as the basis for the TSI calculation.

Plant studies have long shown that temperature and atmospheric humidity interact in water-loss processes [1], [2]. The biological details are different for fungi, but the atmospheric physics of vapour pressure still applies to a moist mushroom surface. This is why VPD is useful as an intermediate environmental measure. The important point is not to treat a plant equation as direct proof of mushroom

physiology; rather, the project uses VPD as a physically meaningful way to combine two environmental variables before calibration against mushroom-room behaviour.

The project-defined TSI takes the calculated VPD and scales it relative to lower and upper VPD boundaries. It also includes an empirical airflow weighting term inherited from the group model. This produces a percentage-like stress score that can be compared with a safety threshold. The benefit for the farmer is that the system can show one understandable stress signal instead of expecting the farmer to interpret two raw sensor values.

2.4 IoT monitoring and environmental control in mushroom cultivation

IoT-based mushroom monitoring has been studied in Sri Lanka and other countries. Wireless sensor nodes can collect temperature and humidity continuously and send the values to a remote application [3], [7], [8]. Other researchers have added fuzzy logic or threshold-based control so that fans, heaters or humidifiers can be operated automatically [4]–[6], [9]. These studies show that the basic hardware problem is already solvable.

The paper by Permata et al. designed an automatic temperature and humidity measuring device using a DHT22 sensor, Arduino Uno, fan, heater and LCD [10]. It is a useful reference because it shows a complete sensor-to-control flow for an oyster mushroom room. However, its main output is local device control and LCD display. The present individual component goes one step further by using the environmental readings to calculate a combined stress value and by sending a room-wise mobile warning with a clear farmer action.

Several existing systems also use fixed environmental limits. Fixed limits are simple, but they can become too rigid when the same humidity level has a different effect at a different temperature. Fuzzy controllers improve this by considering more than one input, but they still usually focus on device control rather than an explainable stress index for the farmer. The TSI approach is valuable because it can be shown as an environmental risk score and tracked over time.

Study / approach	Main input	Main output	Limitation for this thesis
Sri Lankan IoT mushroom monitoring [3]	Temperature, humidity	Remote monitoring	Limited combined stress interpretation
Fuzzy IoT control [4], [5]	Temperature, humidity	Automatic control	Focuses on control rules, not farmer-facing stress index
Monitoring and control systems [6], [7]	Temperature, humidity	Sensor display / control	Often fixed thresholds
Home mushroom IoT system [8]	Environmental sensors	Monitoring and control	General environmental control
Climate-control system [9]	Temperature, humidity	Automated climate control	Does not centre on room-wise TSI warning
Permata et al. [10]	DHT22 temperature and RH	Fan/heater + LCD	Local display; no combined predictive stress score
Proposed individual component	Temperature, RH + configured airflow factor	VPD, TSI, trend warning, mobile recommendation	Needs further field calibration of low/moderate boundary

Table 2.2 Comparison of related monitoring and control approaches.

2.5 Gap synthesis and conceptual framework

The review shows that the main gap is not the absence of temperature or humidity sensors. Those sensors are already widely used. The gap is the step between measurement and useful farmer decision. Existing systems often show separate values, apply fixed thresholds, or directly switch equipment. They provide less support for explaining the combined moisture condition and less support for warning the farmer that stress is developing before a critical limit is crossed.

The conceptual framework of this thesis therefore has six steps. First, the room environment is measured. Second, temperature and RH are combined through VPD. Third, VPD is normalized into TSI. Fourth, the current TSI is compared with configured warning and safety limits. Fifth, recent TSI behaviour is checked to identify whether the condition is stable or moving toward risk. Finally, the result is converted into a simple mobile message that says what is happening and what the farmer should do next.

This framework also supports local adaptation. The environmental targets can be changed by variety and growth stage, while the high-stress safety limit remains based on the group prototype until more field data are available. This makes the

system more appropriate for Sri Lankan mushroom farming than a single fixed global range.

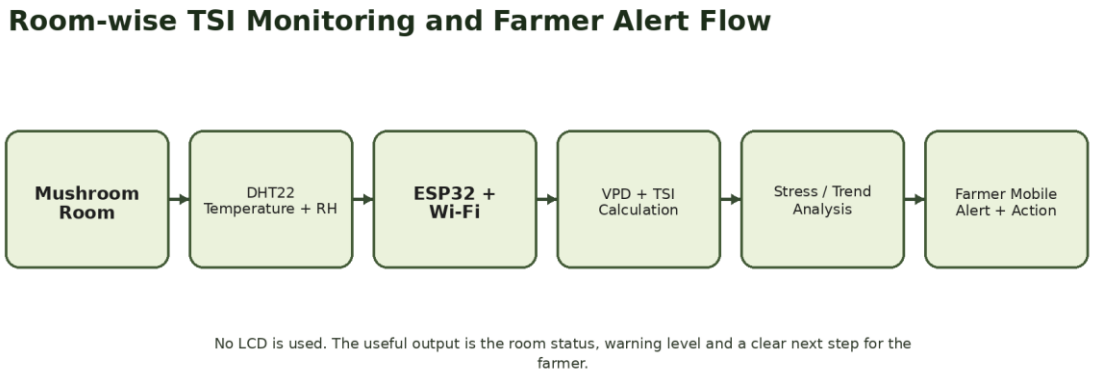


Figure 2.1 Conceptual flow from room conditions to farmer action.

3 RESEARCH METHODOLOGY

3.1 Research design

This research follows a Design Science Research approach because the work is centred on creating and evaluating a practical artefact. The artefact is not only a sensor node. It is the complete TSI decision component: data collection, environmental calculation, warning logic, recommendation logic and integration with the farmer interface. The research moves through repeated cycles of understanding the problem, designing the solution, implementing it, testing it, and improving it.

An iterative software-development approach was used alongside the research process. Sensor communication, backend calculations and mobile output were tested separately before they were connected. This reduced the chance that a problem in one part of the system would hide a problem in another part. The integrated group platform was then used to evaluate response time and user acceptance.

3.2 Data, study materials and environment

The main live data used by this component are temperature and relative humidity. These are captured using DHT22 sensor nodes connected to ESP32 microcontrollers. The group prototype samples environmental data at short intervals and transfers the readings over Wi-Fi using MQTT. The backend stores the time, sensor identity, location and calculated environmental values.

The overall group prototype originally describes localized sensor placement across rack zones. For this individual thesis, the farmer-facing unit is the cultivation room. This does not require the physical sensor architecture to be ignored. Instead, one or more sensor nodes can be assigned to a room, and the room view can use the most appropriate node, a selected reference node, or an aggregation rule depending on deployment size. This change matches the current individual scope without falsely claiming that the original field trial was performed only at room level.

The study also uses local environmental requirements from the Sri Lankan Department of Agriculture. These values are used as context when interpreting the

TSI and when designing recommendations. The group system evaluation provides the available sensor-accuracy, correlation, latency and user-acceptance results used in Chapter 6.

Field	Purpose	Example
timestamp	Keeps the time of each reading	2026-09-30 10:15:20
roomId	Links the reading to a cultivation room	ROOM_01
sensorId	Identifies the physical node	ENV_01
temperatureC	Measured air temperature	24.8
relativehumidity	Measured RH percentage	88.4
vpd	Calculated atmospheric moisture deficit	Calculated value
tsi	Calculated stress index	Calculated value
stresslevel	Farmer-facing interpretation	LOW / MODERATE / HIGH
trend	Recent change in TSI	STABLE / RISING / FALLING
recommendation	Next action for farmer	Maintain / inspect / correct

Table 3.1 Main data fields used by the TSI component.

3.3 Experimental and development procedure

The first step was to confirm the environmental variables that belong to the individual component. Temperature and relative humidity were selected as the live measurements. The group TSI model also includes an airflow weighting term. Because the individual hardware scope does not include a dedicated airflow sensor, this term is treated as a configured empirical value unless measured airflow data are available. This keeps the live sensor requirement consistent with the temperature-humidity focus.

The second step was to verify the sensor readings. Temperature and RH readings were compared with reference measurements during system testing. The purpose of this step was to make sure that an error in the sensor would not directly become an error in TSI. Readings were also checked for impossible or obviously invalid values before they were accepted by the backend.

The third step was the environmental calculation. Saturated vapour pressure was calculated from temperature, then VPD was calculated using RH. The TSI formula normalized VPD between configured lower and upper boundaries and applied the empirical airflow weighting. This process converts the two sensor measurements into one stress score.

The fourth step was stress interpretation. The group research defines TSI greater than 65 as the safety trigger for high stress. A lower warning boundary is needed to separate low and moderate stress, but that boundary has not been fully validated in the available group paper. Therefore, the warning value is configurable. Low stress means the current TSI is below the warning band. Moderate stress means the TSI is approaching the high-stress limit or the trend is rising. High stress means TSI is above the safety threshold and immediate action is required.

The fifth step was early-warning trend analysis. A short rolling window of recent TSI readings is used to determine whether the value is stable, rising or falling. The current thesis treats this as an explainable trend-based prediction rather than claiming a separately trained machine-learning forecasting model. If recent TSI values are rising and the estimated near-term direction is toward the safety limit, the system can send an early warning before the high threshold is reached.

The final step was farmer communication. Instead of showing only a numeric TSI, the mobile output combines room name, current temperature, RH, TSI, stress level and a short next action. This is important because the farmer needs a decision, not only a calculation.

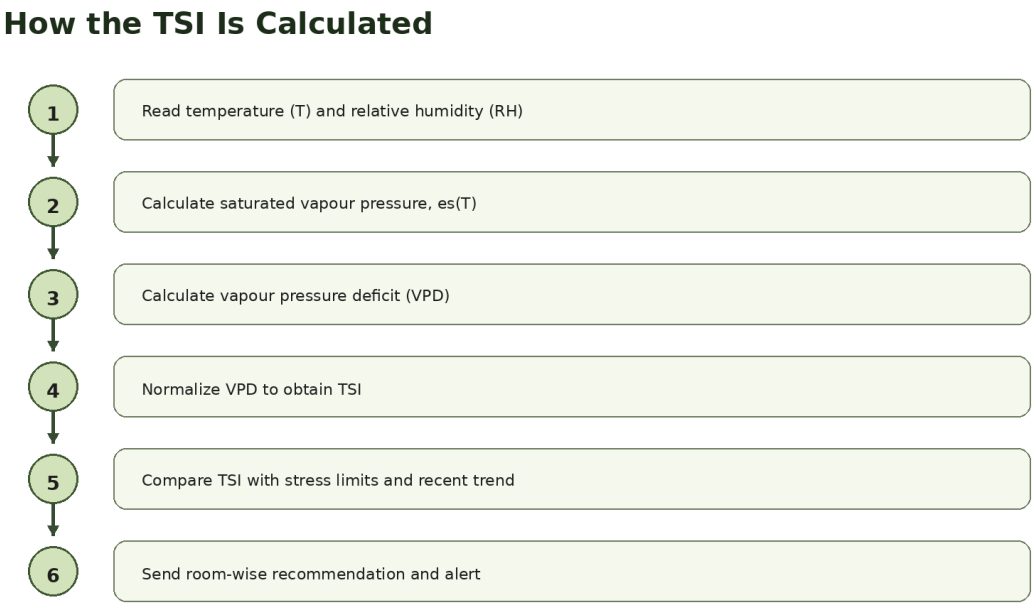


Figure 3.1 TSI calculation and decision procedure.

3.4 Measures and evaluation metrics

The component was evaluated using measures that answer both the research and product questions. Table 3.2 lists the main measures. Some measures come from the integrated group evaluation because the TSI component operates inside the shared platform.

Measure	Why it is important	Available evidence
Temperature accuracy	TSI depends on correct temperature input	Approximately ±0.3 °C in group evaluation
RH accuracy	TSI depends strongly on RH	Approximately ±2.1% RH
Temperature–TSI correlation	Checks whether TSI responds to thermal changes	r = 0.92
RH–TSI correlation	Checks whether TSI responds to humidity changes	r = -0.99
Temperature–RH correlation	Describes environmental interaction	r = -0.90
Actuation/response latency	Shows whether the system can react quickly	Average 4.2 s
Farmer acceptance	Shows whether stress control is useful	Mean 4.73/5; 94.6% satisfaction
Prediction accuracy	Would measure forecast error	Not separately reported; should be added with

Measure	Why it is important	Available evidence
		labelled forecast data

Table 3.2 Evaluation measures used in the study.

3.5 Data analysis

Descriptive statistics were used to check the range and behaviour of the environmental readings. Correlation analysis was used to examine how temperature and RH changed in relation to TSI. Pearson correlation coefficients reported in the group study provide the main quantitative evidence for this relationship.

The high positive correlation between temperature and TSI means that higher temperature tended to be associated with higher calculated stress in the observed data. The strong negative correlation between RH and TSI means that lower humidity tended to increase stress. These relationships are consistent with the VPD calculation, because warmer and drier air creates a larger vapour-pressure difference.

For the early-warning function, the important analysis is not only the current value but also the direction of recent values. A rising TSI sequence indicates that the room is moving toward a more stressful state. Future work should evaluate forecast error using a defined prediction horizon and metrics such as Mean Absolute Error once a sufficiently large labelled time-series dataset is available.

3.6 Ethics, privacy, security and responsible AI

This component collects environmental data rather than sensitive medical or personal data. However, farm names, room identifiers and user-account information can still identify a business or user. Access to the mobile system should therefore be authenticated, and farm data should be visible only to authorised users. Transport security should be used for internet-facing APIs.

Farmer participation in user testing should be voluntary, and participants should understand what is being evaluated. If names or contact details are recorded, they should not be published in the thesis without permission. Reported UAT results should be presented as group-level values.

The TSI warning is a decision-support recommendation. It should not be presented as an infallible biological diagnosis. The system must explain the environmental reason for the alert so that the farmer can inspect the room and use practical judgement. Generative AI is not required to calculate TSI, and no AI model should be allowed to override the validated environmental rules without evidence.

3.7 Reproducibility and research artefacts

A reproducible version of this component should include the sensor configuration, ESP32 firmware, MQTT topic structure, backend calculation code, TSI configuration values, database fields, test cases and example input/output data. Version numbers and configuration files should be stored together with the source code so that another student can repeat the calculation using the same inputs.

The thesis appendices provide a sample data-recording sheet, a worked calculation format, example alert messages and a test-case checklist. Before final submission, the exact repository location, software versions and any deployment-specific configuration used in the final experiment should be added to the project evidence.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements were derived from the research problem and from practical farmer use. The component must be simple enough to understand, but the environmental calculation must remain transparent and testable.

ID	Requirement	Type
FR-01	Read temperature and RH from the environmental sensor node.	Functional
FR-02	Associate each reading with a room and timestamp.	Functional
FR-03	Calculate saturated vapour pressure and VPD.	Functional
FR-04	Calculate TSI using configured boundaries and airflow factor.	Functional
FR-05	Classify the condition for farmer communication as low, moderate or high stress.	Functional
FR-06	Check recent TSI trend and identify rising risk.	Functional
FR-07	Send a room-wise mobile alert with a clear next action.	Functional
FR-08	Store readings and calculated results for later analysis.	Functional
NFR-01	Process new readings with low enough delay for practical intervention.	Non-functional
NFR-02	Keep calculations deterministic and explainable.	Non-functional
NFR-03	Allow thresholds and variety/stage settings to be configured.	Non-functional
NFR-04	Protect farm/user data using authenticated access.	Non-functional

Table 4.1 Functional and non-functional requirements.

4.2 Component architecture within the complete solution

The component has five practical layers. The sensing layer contains the DHT22 and ESP32. The communication layer transfers readings through Wi-Fi and MQTT. The processing layer validates the reading and calculates VPD, TSI and trend. The persistence layer stores time-series results. The presentation layer shows the farmer the room status and recommendation.

The overall group project contains additional services for disease detection, growth-stage analysis and CO2 stress. Those services may share the same mobile

interface and database, but they are separate from the TSI calculation. Keeping the services separate makes it easier to test each research contribution and prevents one model from changing the meaning of another component's result.

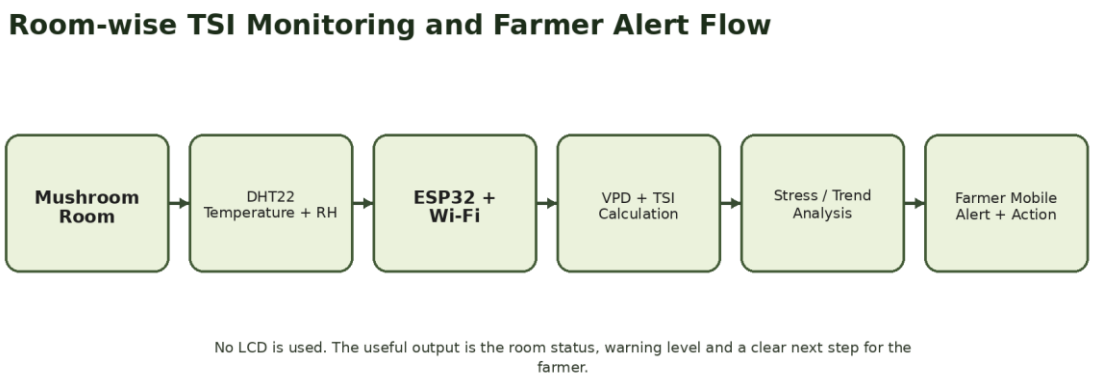


Figure 4.1 Room-wise TSI monitoring and farmer alert architecture.

4.3 Design alternatives and decisions

Design question	Alternative considered	Selected decision	Reason
Display method	Local LCD	Mobile room-wise status	Farmer can see alerts remotely; no LCD dependency
Environmental logic	Separate temp/RH thresholds	VPD + TSI	Combines the effect of both readings
Stress levels	Only normal/critical	Low / moderate / high	Provides an early-warning stage
Low–moderate boundary	Hard-code an invented number	Configurable warning threshold	Available evidence validates high limit, not lower boundary
Prediction	Complex ML model without enough labels	Recent-trend early warning	Simple, explainable and does not overclaim accuracy
Farmer scope	Only rack label	Room-wise presentation	Matches current individual requirement and farmer workflow
Airflow factor	Require new live sensor	Use configured empirical factor unless sensor available	Keeps live inputs temperature + RH

Table 4.2 Main design decisions and reasons.

4.4 Detailed design

The backend receives a message containing sensor identity, room identity, temperature and RH. The message is checked before use. A missing room ID, invalid temperature or impossible RH value is rejected or marked as invalid. Valid readings are then passed to the TSI calculation service.

The first calculation is saturated vapour pressure. The group research uses Tetens' equation, shown in Equation 4.1. Temperature T is measured in degrees Celsius and the result is expressed in kilopascals.

$$e_s(T) = 0.61078 \times \exp(17.27T / (T + 237.3)) \quad (4.1)$$

VPD is then calculated from saturated vapour pressure and relative humidity as shown in Equation 4.2. A higher VPD means the surrounding air has a stronger drying demand.

$$VPD = e_s(T) \times (1 - RH / 100) \quad (4.2)$$

The project-defined TSI normalizes VPD between configured lower and upper boundaries and applies an airflow weighting term. Equation 4.3 follows the group research paper. In this individual component, the airflow value v is a configured empirical parameter unless measured airflow data are available.

$$TSI = [(VPD - VPD_{min}) / (VPD_{max} - VPD_{min})] \times (1 + \gamma v) \times 100 \quad (4.3)$$

The group prototype treats TSI greater than 65 as high stress. The system therefore uses 65 as the current safety threshold. A warning threshold below 65 is stored as configuration rather than presented as a scientifically fixed value. This

allows later farm trials to calibrate the low-to-moderate boundary without changing the application code.

The recommendation engine uses both stress level and the direction of the environmental change. A low and stable condition produces a “maintain” message. A moderate or rising condition produces an “inspect and adjust” warning. A high condition produces an immediate corrective-action alert. The recommendation can mention the reason. For example, if temperature is high and RH is falling, the message can advise the farmer to check room cooling and humidification rather than displaying only “TSI High”.

A room status record is updated each time a valid reading is processed. Historical records are kept so the farmer can view recent trends and so the research team can later evaluate how stress develops over time.

Farmer-facing state	Meaning	Recommended message
Low stress	Current TSI is below the configured warning band and trend is stable.	Conditions are suitable. Continue normal monitoring.
Moderate stress	TSI is approaching the safety limit or rising quickly.	Warning: stress is increasing. Inspect the room and check temperature/humidity control.
High stress	TSI is above the group safety limit (>65).	Critical: take corrective action now. Check humidification and excessive temperature, then confirm the room condition.

4.5 Standards and professional practice

The design follows common IoT and software-engineering practice by separating sensing, communication, processing, storage and presentation. MQTT is suitable for small telemetry messages, while HTTPS should be used for mobile/API communication. Timestamps should use a consistent time zone and storage format so that trend analysis is correct.

Professional practice also requires traceability. Every alert should be linked to the measurements that caused it. This makes the decision explainable and helps debugging. Configuration changes, such as warning limits or variety/stage selection, should be recorded so that later results can be interpreted correctly.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development environment and technology stack

The environmental node uses a DHT22 sensor connected to an ESP32 microcontroller. The ESP32 provides the processing and Wi-Fi capability needed to sample the sensor and publish readings. MQTT is used for lightweight telemetry transfer. The group architecture uses a cloud-backed backend, MongoDB for persistence and a React Native mobile application for live information.

The TSI calculation is implemented as backend logic so that the same calculation is used for all connected rooms. Keeping the calculation in the backend also makes configuration easier to update and prevents different mobile devices from calculating different results. The mobile application receives the processed result rather than reproducing the full environmental formula locally.

5.2 Component implementation

The ESP32 firmware initializes the DHT22, reads temperature and RH, checks whether a valid reading is available, adds the sensor identifier and sends the payload to the MQTT broker. A short sampling interval allows the backend to detect changes quickly. The group paper reports a ten-second sampling interval for the prototype.

When the backend receives a reading, it validates the payload and stores the raw values. It then calculates saturated vapour pressure, VPD and TSI using the configured boundaries. The current TSI is appended to a short recent-history window. The trend is classified as rising, stable or falling. The backend then updates the current room status and decides whether an alert should be sent.

The alert message is deliberately simple. It contains the room name, current temperature, current RH, TSI level and one next action. The detailed numeric TSI may be available on the dashboard, but the notification text is written for a farmer who may not know the formula.

Simplified processing logic
1. Receive roomId, temperature and RH.
2. Validate the reading.
3. Calculate e_s(T).
4. Calculate VPD.
5. Calculate TSI.
6. Update recent TSI history and trend.
7. If TSI > 65, set HIGH stress and send critical action.
8. Else if TSI is in the configured warning band or rising toward the limit, set MODERATE stress.
9. Else set LOW stress.
10. Store the result and update the mobile room status.

5.3 Integration with the other components

The TSI component publishes its result to the shared backend and mobile application. Other components can display their own disease, growth-stage or CO2 information on the same room or farm screen. The integration point is the shared identity of the farm, room, rack or sensor location, not a merged algorithm.

This separation is important for the individual thesis. A disease alert must not be reported as a TSI result, and a CO2 warning must not change the TSI formula. The farmer may see several warnings in one application, but each warning keeps its own source and explanation.

5.4 Verification and software testing

Testing was carried out at several levels. Sensor verification checked whether measured temperature and RH were close to reference values. Unit tests checked the VPD and TSI calculation using known inputs. Boundary tests checked the behaviour below and above the high-stress threshold. Integration tests checked the path from sensor message to stored result and mobile output. User testing examined whether the automation and stress-control feature was understandable and useful.

The early-warning logic also needs sequence-based tests. A single value may be moderate, but a steadily rising sequence is more important than one isolated reading. Therefore, test cases include stable, rising and falling TSI series.

Test ID	Scenario	Expected result
T01	Valid temperature and RH payload	Reading accepted and stored
T02	RH outside 0–100%	Reading rejected or flagged invalid
T03	Known T and RH calculation case	Expected VPD and TSI returned within numeric tolerance
T04	TSI below warning band and stable	LOW state; no urgent alert
T05	TSI below 65 but rising toward threshold	MODERATE early warning
T06	TSI > 65	HIGH state; critical alert and control action
T07	TSI falling after intervention	Stress state reduces when configured conditions are met
T08	Sensor/network interruption	System records stale/offline condition instead of using old value as live
T09	Two rooms reporting simultaneously	Each room keeps separate status/history
T10	Mobile alert	Alert shows room, condition and recommended next step

Table 5.1 Verification and test cases.

5.5 Deployment and operation

In operation, each sensor node is assigned to a known physical location. The farmer-facing dashboard groups readings by cultivation room. Larger rooms can use more than one node if the environment is not uniform. The system should show when a sensor is offline so that an old reading is not mistaken for a current safe condition.

The recommended deployment process is to install and verify the sensor, confirm the room and variety/stage settings, observe the data for a calibration period, and then enable warning notifications. The high-stress safety limit is based on the group prototype, while lower warning settings should be reviewed using local farm data. This staged approach reduces false alerts and makes the system easier for farmers to trust.

6 RESULTS AND VALIDATION

6.1 Research results

The available quantitative results come from the environmental subsystem inside the integrated group evaluation. They are used here because they directly measure the inputs and outputs of this individual TSI component. Results from disease detection and growth-stage classification are not treated as evidence for this thesis.

The sensor comparison reported approximately ±0.3 °C temperature accuracy and ±2.1% RH accuracy. These values are within the target range stated in the group evaluation and indicate that the environmental inputs were stable enough for the TSI calculation. The study also found a strong inverse relationship between temperature and RH ($r = -0.90$), which is expected in a closed cultivation environment when warmer conditions are associated with lower relative humidity.

TSI had a strong positive relationship with temperature ($r = 0.92$) and a very strong negative relationship with RH ($r = -0.99$). This confirms that the calculated index behaves in the expected direction: warmer and drier conditions produce higher stress values.

Result	Reported value	Interpretation
Temperature accuracy	±0.3 °C	Suitable for environmental monitoring in the group prototype
Relative-humidity accuracy	±2.1% RH	Suitable for TSI input within the tested setup
Temperature vs RH	r = -0.90	Strong inverse relationship
Temperature vs TSI	r = 0.92	Higher temperature associated with higher TSI
RH vs TSI	r = -0.99	Lower humidity strongly associated with higher TSI
Average response/actuation latency	4.2 s	Fast enough for near-real-time environmental response

Table 6.1 Reported environmental and TSI results from the integrated group evaluation [14].

6.2 Product and technical results

The integrated system achieved an average response latency of 4.2 seconds for the TSI/telemetry control path. This is faster than the five-to-ten-second target window stated in the group paper. It shows that the system can move from environmental measurement to a control or warning decision without a long delay.

The group paper also reports that closed-loop TSI-based control maintained the tested environmental conditions within the broad prototype range of 20–26 °C and 80–90% RH. This result supports the technical value of connecting the stress calculation to an action. However, the broad range should not be interpreted as the only correct range for every Sri Lankan variety and stage. The Department of Agriculture values in Table 2.1 remain the better local reference for configuration.

6.3 User or stakeholder validation

The integrated platform was evaluated in a four-week field trial across three commercial mushroom farms with 15 participants, including experienced and novice growers. The user study measured several parts of the overall platform. The result most directly related to this individual component is the category "Automation & Stress Control".

That category received a mean score of 4.73 out of 5 and a satisfaction rate of 94.6%. This suggests that growers found the TSI-triggered environmental support useful. The thesis does not attribute the overall 97.4% operational-value score only to this component because that score also reflects benefits from other parts of the group system.

UAT category	Assessed feature	Mean score	Satisfaction
Automation & Stress Control	Effectiveness of automated misting/stress-control response triggered by TSI thresholds	4.73 / 5	94.6%

Table 6.2 User validation result directly related to the TSI component [14].

6.4 Integration results

The environmental component successfully operated inside the shared platform and was able to exchange data with the common persistence and mobile layers. This demonstrates that the component can remain independent at the algorithm level while still participating in one farmer application.

The integrated result is important because farmers should not need a separate application for each research component. At the same time, clear labels are required so that a temperature/humidity stress warning is not confused with a disease or CO2 warning. The shared location identifier supports this separation.

6.5 Summary of findings

Objective	Evidence / result	Status
Collect continuous temperature and RH	DHT22 + ESP32 telemetry operated in integrated prototype	Achieved
Calculate VPD and TSI	TSI used in live environmental control pipeline	Achieved
Provide high-stress decision	Group safety trigger TSI > 65 used	Achieved
Provide low/moderate/high farmer messages	Design and mobile logic support three levels; lower warning boundary remains configurable	Partly calibrated
Provide early warning from recent trend	Trend-based logic designed and testable; separate forecast error not reported	Needs further quantitative validation
Give farmer mobile recommendation	Integrated stress-control function received high UAT rating	Achieved
Evaluate environmental subsystem	Accuracy, correlation, latency and UAT results available	Achieved

Table 6.3 Objective-to-result summary.

7 DISCUSSION

7.1 Interpretation of findings

The results show that the TSI calculation is strongly driven by the two environmental variables it was designed to combine. The positive temperature–TSI relationship and negative RH–TSI relationship are consistent with the VPD formula. This gives confidence that the index is behaving logically within the tested range.

The result also explains why separate thresholds can be weak. If temperature rises while RH falls, both changes push VPD in the same stressful direction. A farmer looking at two numbers must mentally combine them. The TSI performs this combination consistently and allows the mobile application to show one clear environmental warning.

The 4.2-second average response is practically important. Moisture conditions can change quickly around fruiting bodies, especially when ventilation or warm outside air affects the room. A short processing delay means the system can warn or act while the condition is still developing.

7.2 Comparison with literature baselines and existing solutions

The related literature confirms that DHT-class sensors, microcontrollers and IoT communication are common in mushroom monitoring [3], [6], [7], [10]. Therefore, the novelty of this thesis is not simply connecting a DHT22 to an ESP32. The contribution is the decision layer built on top of those readings.

The automatic device described by Permata et al. uses temperature limits to switch heating and fans and displays the result on an LCD [10]. Fuzzy-control studies use more flexible rules [4], [5]. The present component differs by creating a VPD-based TSI, tracking the stress direction, and sending a farmer-facing recommendation remotely. This is closer to decision support than simple measurement.

The Sri Lankan Department of Agriculture guidance also highlights a limitation of generic environmental control. Different varieties and stages have

different suitable conditions [11]. This supports the decision to keep thresholds configurable rather than presenting one fixed global setting as correct for all farms.

7.3 Research contribution

The research contribution is a transparent method for turning common environmental measurements into a single moisture-stress interpretation for oyster mushroom cultivation. The component links temperature and RH through VPD, maps VPD to TSI, adds recent-trend interpretation and then converts the result into a simple warning level.

A second contribution is the careful separation between what is validated and what still needs calibration. The high-stress threshold is taken from the group research, while the low-to-moderate boundary is not invented. This improves the scientific quality of the thesis because it avoids presenting an unsupported number as a proven biological threshold.

A third contribution is the Sri Lankan contextualization. The thesis uses local variety- and stage-specific guidance to show how the TSI service can be configured for real cultivation practice rather than relying only on a broad test range.

7.4 Product contribution

The product contribution is the farmer-facing decision flow. A farmer does not need to understand VPD equations. The application translates the calculation into room status and a next step. This reduces the gap between technical sensing and practical crop management.

The component also removes the dependence on an LCD attached to the device. A mobile warning is more useful when the farmer is outside the growing room. Room-wise grouping makes the information easier to relate to farm operations, especially when several rooms are managed at the same time.

7.5 Limitations and threats to validity

The first limitation is that the TSI model is a project-defined index and not a universally validated physiological standard for all Pleurotus varieties. More

biological validation is needed to connect specific TSI values to measurable mushroom quality, water loss or yield outcomes.

The second limitation is the low-to-moderate threshold. The group paper provides the high-stress safety limit but not a validated warning boundary. The current design therefore uses a configurable warning band. This is correct for a prototype but should be calibrated through longer field trials.

The third limitation is prediction evaluation. The component uses recent TSI trend for early warning, but the available group results do not report a separate forecast accuracy metric. Future experiments should define a prediction horizon and compare predicted TSI with observed TSI.

The fourth limitation is sensor placement. The overall prototype uses localized rack-zone measurements, while the current farmer view is room-wise. A single sensor may not represent a large room with strong spatial differences. Multi-node aggregation needs further study.

The fifth limitation is that the TSI formula includes an airflow weighting term. If airflow is not measured live, a configured value is an approximation. Future work can add an airflow sensor or recalibrate the coefficient for different room designs.

Finally, the available UAT result comes from the integrated platform. Although the “Automation & Stress Control” category directly relates to TSI, user experience may still be influenced by the rest of the application.

7.6 Implications

For farmers, the main implication is that simple environmental sensors can provide more useful information when the data are interpreted rather than only displayed. A room warning that explains the reason and next action can support less-experienced growers and reduce the need for constant manual checking.

For researchers, the study shows the need to validate combined environmental indices directly against mushroom outcomes. Future work should collect larger time-series datasets together with growth, weight, quality and yield observations. This would allow the warning boundaries to be based on evidence instead of only configuration.

For product developers, the study suggests that environmental recommendations should be configurable by variety and stage and should remain explainable. Farmers are more likely to trust a warning when the application shows the sensor values and the reason for the recommendation.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The main target users are small- and medium-scale oyster mushroom farmers who need simple environmental monitoring without expensive industrial control systems. The value proposition is straightforward: measure the room continuously, identify moisture stress early, and tell the farmer what action to take.

The component can also support training centres, farm advisors and larger farms that manage several cultivation rooms. Because the calculation is centralized, new rooms can be added without changing the core logic.

8.2 Market and competitive position

Many low-cost monitoring products can show temperature and humidity, so a commercial product should not compete only as another sensor display. The differentiating feature is the interpretation layer: Sri Lankan variety/stage settings, TSI-based stress level, recent trend and practical farmer recommendation.

The product should remain low-cost by using commonly available sensors and microcontrollers. It can be positioned between a basic thermometer/hygrometer and a full industrial climate-control system.

8.3 Business or sustainability model

A practical model could combine a one-time hardware cost with a low monthly software fee for cloud storage, alerts and multi-room monitoring. An offline or local-network version could also be offered where internet access is unreliable. Farmer cooperatives or agricultural extension programmes could deploy the system to several growers at lower unit cost.

Long-term sustainability depends on keeping maintenance simple. Sensors should be replaceable, configuration should be backed up, and the software should continue to work even when a particular mobile device is changed.

8.4 Scalability, intellectual property and compliance

The software can scale by storing each reading with farm and room identifiers and by processing telemetry as independent events. More sensor nodes increase data volume, but the TSI calculation itself is lightweight. The main scaling concerns are message volume, database retention and alert rate.

The TSI equation and general VPD methods are based on published knowledge and the group research. Intellectual property may therefore lie more in the complete implementation, local calibration dataset, recommendation rules, user interface and deployment process than in claiming ownership of the physical formula itself.

Any commercial deployment should follow electrical safety, data-protection and wireless-device requirements applicable in Sri Lanka. The system should also avoid giving guarantees about crop yield unless those claims are supported by field evidence.

8.5 Contribution to the Sustainable Development Goals

The component supports SDG 2 (Zero Hunger) by improving information used in food production, SDG 8 (Decent Work and Economic Growth) by supporting small agricultural businesses, SDG 9 (Industry, Innovation and Infrastructure) through affordable IoT use, and SDG 12 (Responsible Consumption and Production) by helping farmers use water and environmental control more carefully.

These contributions are indirect and should not be overstated. The system supports better decisions; the actual sustainability benefit depends on how farmers use the recommendations and how much waste, water or crop loss is reduced in future field studies.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis focused on one clearly defined part of the R26-ISE-002 project: the temperature- and humidity-based TSI component for oyster mushroom cultivation. The work addressed a practical problem found in many monitoring systems: raw sensor values are available, but the farmer still has to decide what the combination means and what action should be taken.

The proposed component combines temperature and RH through VPD and the project-defined TSI, checks current stress and recent trend, and presents the result as a room-wise farmer warning. The group evaluation showed that the environmental telemetry was accurate within the tested setup, the TSI responded strongly to changes in temperature and RH, and the control path reacted within a few seconds. Farmers also rated the automation and stress-control feature positively.

The most important outcome is therefore not the sensor itself. It is the conversion of environmental data into understandable decision support. The research also identifies where further validation is still required, especially the lower warning boundary, airflow calibration, and quantitative forecast accuracy.

9.2 Achievement of objectives

1. Continuous temperature and RH collection was implemented using DHT22 and ESP32 telemetry.
2. VPD and TSI calculations were integrated into the environmental processing flow.
3. The high-stress safety decision uses the threshold reported by the group research.
4. Farmer communication supports low, moderate and high messages, while the lower boundary remains configurable until calibrated.
5. Trend-based early warning is included as an explainable prediction mechanism, but separate prediction-accuracy evaluation remains future work.
6. Room-wise mobile alerts and recommendations replace the need for an LCD.

7. Available validation includes sensor accuracy, TSI correlations, response latency and UAT evidence.

9.3 Individual research and product contributions

The individual research contribution is the use of a combined temperature-humidity stress interpretation, rather than separate raw thresholds, together with clear acknowledgement of the limits of the current calibration. The product contribution is the room-wise farmer decision flow that turns the calculation into a simple warning and next action.

The thesis also contributes a local configuration approach based on Sri Lankan Department of Agriculture guidance. This makes it possible to adapt the environmental interpretation to different oyster mushroom varieties and growth stages.

9.4 Future directions

Future work should collect longer time-series data from multiple Sri Lankan farms and record mushroom quality, weight, yield and visible moisture-stress outcomes alongside the environmental data. These labels can be used to calibrate the low and moderate TSI boundaries and test whether the current high threshold is suitable across varieties.

A second direction is to evaluate prediction formally. A fixed forecast horizon should be selected, predicted TSI should be compared with observed TSI, and error measures such as MAE or RMSE should be reported. If the simple trend method is not accurate enough, time-series methods can then be tested using the same dataset.

A third direction is to include measured airflow, improve multi-sensor room aggregation, detect sensor drift, and support local/offline alerting when internet connectivity is unavailable. Future versions can also recommend variety- and stage-specific actions automatically after the farmer selects the current crop stage.

REFERENCES

[1] T. Kuwagata et al., “Influence of low air humidity and low root temperature on water uptake, growth and aquaporin expression in rice plants,” Plant and Cell Physiology, vol. 53, no. 8, pp. 1418–1431, 2012.

[2] H. M. Rawson, J. E. Begg, and R. G. Woodward, “The effect of atmospheric humidity on photosynthesis, transpiration and water use efficiency of leaves of several plant species,” Planta, vol. 134, no. 2, pp. 135–140, 1977.

[3] S. Yd et al., “IoT-based Monitoring System for Oyster Mushroom Farms in Sri Lanka,” General Sir John Kotelawala Defence University Repository, 2022.

[4] D. Eka Halim, Yultrisna, and Muhammad, “Temperature and Humidity Control in Oyster Mushroom Cultivation Using the Fuzzy Sugeno Method Integrated with IoT,” Jurnal Kelitbangan, vol. 12, no. 3, 2024.

[5] S. P. Budiarto, A. H. Sumitro, and M. Taufiq, “Intelligent System Design for Oyster Mushroom Cultivation Using Mamdani Fuzzy Inference with Internet of Things (IoT),” International Journal of Engineering Continuity, vol. 2, no. 2, pp. 86–106, 2024.

[6] R. A. Mutaqi, M. Yuliza, Y. Antonisfia, and N. R. Sakinah Nisa, “Design of Monitoring and Control System for Air Temperature and Humidity in Oyster Mushroom Cultivation Room Based on IoT,” International Journal of Wireless & Multimedia Communications, vol. 2, no. 1, 2025.

[7] M. F. A. Hamid, A. H. Ahmad Nizam, F. N. A. Ismail, and M. F. Ismail, “Wireless Temperature and Humidity Monitoring System for Oyster Mushroom House,” Multidisciplinary Applied Research & Innovation, vol. 5, no. 2, pp. 167–171, 2024.

[8] J. L. Chong, K. W. Chew, A. P. Peter, H. Y. Ting, and P. L. Show, “Internet of Things (IoT)-Based Environmental Monitoring and Control System for Home-Based Mushroom Cultivation,” Biosensors, vol. 13, no. 1, p. 98, 2023.

[9] C. J. T. Niog, “IoT-Based Climate Control System for Cultivation of Oyster Mushroom Fruiting Bags,” Kyushu University Repository, 2024.

[10] R. S. Permata, D. N. Ilham, M. M. Risky, M. K. Harahap, and S. Kurniadi, “Design of an automatic temperature and humidity measuring device in the oyster mushroom cultivation room,” PERFECT: Journal of Smart Algorithms, vol. 1, no. 1, 2024, doi: 10.62671/perfect.v1i1.3.

[11] Department of Agriculture Sri Lanka, Horticultural Crops Research and Development Institute, “Mushroom – Oyster mushrooms: climatic requirements/areas suitable for cultivation,” accessed Oct. 6, 2026.

[12] S. T. D. De Silva, N. S. B. M. Atapattu, and K. L. W. Kumara, “The Water Footprint of Oyster mushroom (Pleurotus ostreatus) Cultivation under Small-scale Polybag Farming Conditions in Sri Lanka,” The Journal of Agricultural Sciences – Sri Lanka, vol. 18, no. 2, pp. 172–182, 2023.

[13] R. Rafaya and N. Jeyagowri, “Alternate substrate formula for mushroom cultivation by utilizing coir waste,” Sri Lankan Society for Microbiology Annual Conference and Scientific Sessions, 2024.

[14] R26-ISE-002 Research Group, “Rack-Level Micro-Climate and Growth Optimizer for Oyster Mushroom Farms,” IEEE-format group research manuscript, Sri Lanka Institute of Information Technology, 2026.

GLOSSARY

Term	Simple definition
Airflow factor	A configured value used in the group TSI equation to represent the effect of air movement.
Early warning	A message sent before the high-stress limit is crossed when recent TSI is moving toward risk.
Relative humidity (RH)	The amount of water vapour in the air compared with the maximum the air can hold at the same temperature.
Room-wise monitoring	Presenting environmental condition and alerts by cultivation room.
TSI	The project-defined stress index calculated from VPD, configured boundaries and an airflow weighting term.
Vapour Pressure Deficit (VPD)	The difference between how much moisture the air can hold and how much moisture it currently contains; a measure of atmospheric drying demand.

APPENDICES

Appendix A – Sample environmental data sheet

Date/time	Room	Temp °C	RH %	VPD	TSI	Stress	Trend	Action

Appendix B – Worked TSI calculation format

Use one real sensor reading from your final dataset and complete the fields below. This appendix is intentionally a calculation worksheet so that the final submitted thesis can show one traceable example from raw reading to farmer decision.

Temperature T = _____ °C

Relative humidity RH = _____ %

Configured VPD_min = _____ kPa

Configured VPD_max = _____ kPa

Configured airflow factor v = _____

Configured coefficient γ = _____

e_s(T) = _____

VPD = _____

TSI = _____

Stress level = _____

Recent trend = _____

Farmer recommendation = _____

Appendix C – Example farmer alerts

Low stress example

ROOM 01 – Conditions are suitable. Temperature and humidity are within the current target. Continue normal monitoring.

Moderate stress example

ROOM 01 – Warning: environmental stress is increasing. Please inspect the room and check humidity, misting and temperature control before the condition becomes critical.

High stress example

ROOM 01 – Critical environmental stress. TSI is above the safety limit. Take corrective action now: check humidification and excessive temperature, then confirm the room condition.

Sensor offline example

ROOM 01 – Environmental sensor data are not updating. Check sensor power or network connection. Do not treat the last reading as the current room condition.

Appendix D – Suggested evidence to attach before final submission

The main thesis is complete using the evidence available in the group research paper and current project records. The following items should be added or confirmed before the university submission if they are available. They are evidence items, not new research claims.

- One screenshot of the final mobile room-status screen and one screenshot of a TSI alert.
- A small export of real timestamped temperature, RH and TSI data used in the final demonstration or field test.
- The exact warning-threshold configuration used in the final build and a short note explaining how it was chosen.
- The exact recent-trend / prediction window implemented in source code. If it differs from the description in Chapter 5, update the method section.
- The final firmware, backend and mobile version numbers or repository commit IDs.

- Any sensor calibration sheet or reference-device comparison used to support the reported ±0.3 °C and ±2.1% RH result.
- The participant information/consent or ethics evidence required by the module for the UAT.
- Final supervisor-approved wording for the official thesis title, degree title and submission month.

Appendix E – Academic integrity and AI-use disclosure

If generative AI was used to support language editing, formatting, brainstorming, code explanation or document preparation, list the tool, purpose, representative prompts/workflow and the checks performed. The student remains responsible for verifying all technical facts, equations, citations, results and conclusions against the original project evidence.